18A
CHEMICAL THERMODYNAMICS

UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 6

1. State the second law of thermodynamics.
2. Define entropy (S) and identify the sign of ΔS from chemical equations or changes of state.
3. State the third law of thermodynamics.
4. Calculate ΔS_{rxn} from molar entropies.
5. Compare and contrast ΔS_{univ}, ΔS_{sys}, and ΔS_{surr}.
6. Define Gibbs free energy (G) and describe how the signs of ΔH and ΔS affect the sign of ΔG.
7. Relate the sign of ΔG to the favorability/spontaneity of a reaction.
8. Determine the equilibrium temperature of a reaction.
9. Apply the relationship between $\Delta G°$ and the equilibrium constant (K).

Entropy and Spontaneity

18.1

Much of the work of chemists is to *make predictions* and answer questions about physical and chemical processes.

• What *changes* in a chemical reaction?	Chemical equations
• In what *proportions/quantities*?	Stoichiometry
• How *fast/slow* does a reaction happen?	Kinetics and k
• To *what extent* does a reaction occur?	Equilibrium and K
• Is *heat* absorbed or released during the reaction?	Enthalpy: ΔH_{rxn} and q_{rxn}

The focus of this chapter (thermodynamics) is:

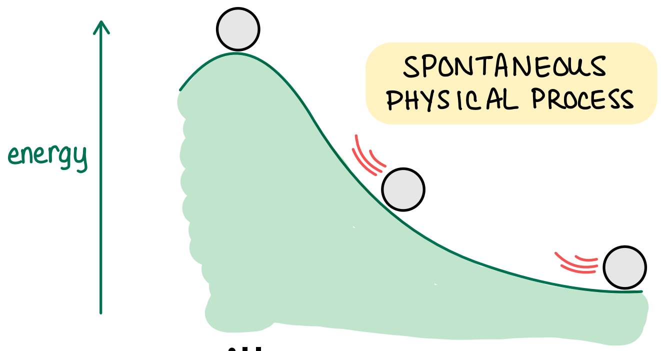
• Does a reaction/process happen ***spontaneously***?	Thermodynamics

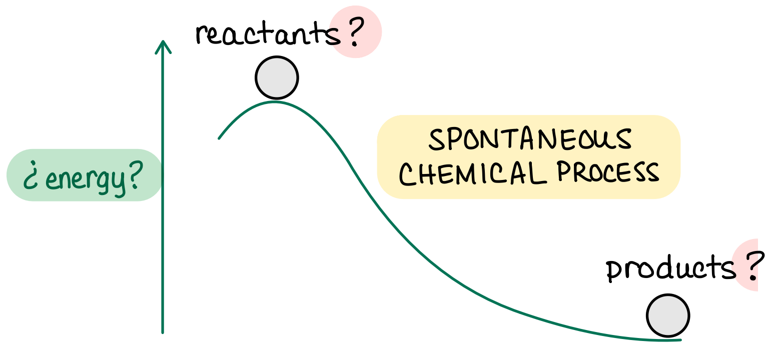
Entropy and Spontaneity

18.1

Spontaneous processes: those that occur, under the specified conditions, ***without* the *continuous* supply of an *external* force/influence**.

- Spontaneity does *not* relate to time.
- Some thermodynamically spontaneous process occur so slowly they they cannot be observed in our lifetime or even several lifetimes.
- Both endothermic ($\Delta H_{\mathrm{rxn}} > 0$) and exothermic ($\Delta H_{\mathrm{rxn}} < 0$) processes can be spontaneous.





Nonspontaneous processes will never occur on their own no matter how long you wait.

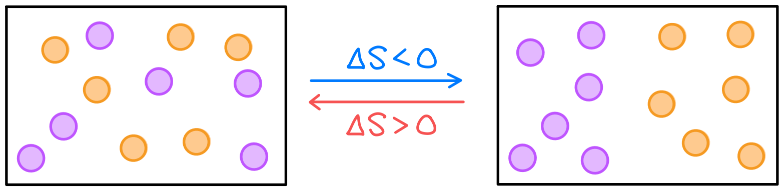
Entropy and Spontaneity

18.1

Entropy (S) is a measure of the degree of disorder (chaos or randomness) in a system.

- Entropy describes the *distribution* of energy within a system.
- When $\Delta S < 0$ $\longrightarrow$ final state less disordered (or more ordered)
- When $\Delta S > 0$ $\longrightarrow$ final state more disordered (or less ordered)

more disordered (or less ordered)



less disordered (or more ordered)

Second Law of Thermodynamics

- A *spontaneous process* is one that results in an overall *increase* in the entropy (disorder) of the universe.

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$$

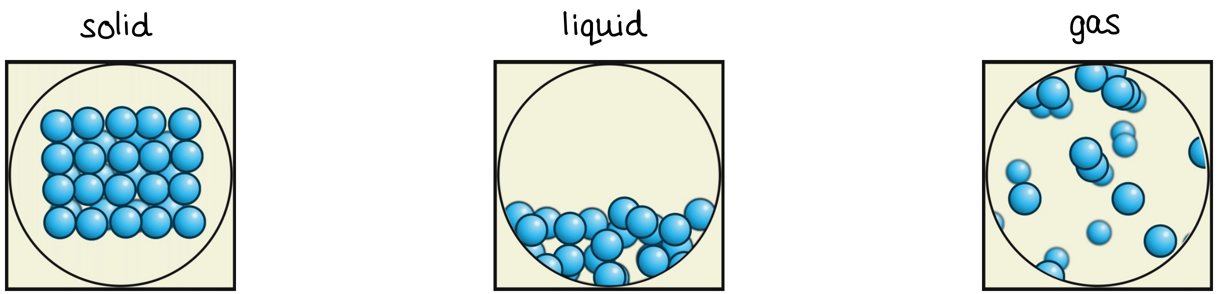
Entropy and Spontaneity

18.1

Factors that *affect the entropy* of a system:

1. Quantity or amount of substance greater quantity, greater S
2. State of matter

$S_{gases} \gg S_{liquids} > S_{solids}$



3. Gases if rxn consumes net moles gas $\rightarrow \Delta n_{gas} < 0 \rightarrow \Delta S < 0$
   if rxn produces net moles gas $\rightarrow \Delta n_{gas} > 0 \rightarrow \Delta S > 0$
4. Temperature higher temp, greater S

EXAMPLE PROBLEM

Learning Objective(s): 18.2

For each process, predict the sign of ΔS_{rxn}.

Reaction or Process		Sign of ΔS_{rxn}	
$Ca_3(PO_4)_2(s) \rightarrow 3\,Ca^{3+}(aq) + 2\,PO_4^{3-}(aq)$	compare phases and quantities	$\Delta S_{rxn} < 0$	$\Delta S_{rxn} > 0$
$I_2(s) \rightarrow I_2(g)$	compare phases solid → gas → $\Delta S > 0$	$\Delta S_{rxn} < 0$	$\Delta S_{rxn} > 0$
$2\,SO_3(g) \rightleftharpoons 2\,SO_2(g) + O_2(g)$	$\Delta n_{gas} = 3 - 2 = 1 > 0 \rightarrow \Delta S > 0$	$\Delta S_{rxn} < 0$	$\Delta S_{rxn} > 0$
Freezing of water	liquid → solid → $\Delta S < 0$	$\Delta S_{rxn} < 0$	$\Delta S_{rxn} > 0$
Condensation	gas → liquid → $\Delta S < 0$	$\Delta S_{rxn} < 0$	$\Delta S_{rxn} > 0$

for phases: $S_{gases} \gg S_{liquids} > S_{solids}$

IN-CLASS QUESTION 1

Learning Objective(s): 18.2

greatest increase in disorder

For which process do you expect ΔS_{rxn} to be the *most positive*?

- ☒ $O_2(g) + 2\ H_2(g) \rightarrow 2\ H_2O(g)$ $\Delta n_{gas} = 2 - 3 = -1 < 0 \rightarrow \Delta S < 0$
- ☒ $H_2O(\ell) \rightarrow H_2O(s)$ liquid $\rightarrow$ solid $\rightarrow \Delta S < 0$
- ☒ $NH_3(g) + HCl(g) \rightarrow NH_4Cl(g)$ $\Delta n_{gas} = 1 - 2 = -1 < 0 \rightarrow \Delta S < 0$
- ☑ $2\ NH_4NO_3(s) \rightarrow 2\ N_2(g) + O_2(g) + 4\ H_2O(g)$ $\Delta n_{gas} = 7 - 0 = 7 > 0 \rightarrow \Delta S > 0$
- ☐ $N_2O_4(g) \rightleftharpoons 2\ NO_2(g)$ $\Delta n_{gas} = 2 - 1 = 1 > 0 \rightarrow \Delta S > 0$

Both have $\Delta S > 0$ because $\Delta n_{gas} > 0$, but $\Delta n_{gas} = 7$ much greater disorder!

Entropy Changes

18.2

Third Law of Thermodynamics

- The absolute entropy of a pure, *perfectly ordered*, crystalline (solid) substance at absolute zero (0 Kelvin) is zero, $S = 0$. i.e. no disorder!
- At absolute zero (0 K), there is no thermal energy (no motion).
- The third law provides a reference, a "zero point" ($S = 0$)!

Standard absolute molar entropies ($S°$)

- The absolute values of entropy (S) based on the "zero point" established by the third law of thermodynamics.
- Recall that the ° symbol indicates standard conditions: gases → 1 atm , aq solns → 1 M , T = 25 °C
- This is unlike enthalpy where *only changes* (ΔH) could be measured.

Entropy Changes

18.2

Truncated table of standard molar entropies at 298 K ($S°$)

- These tables provided when needed, just like ΔH_f^o.
- Greater values of $S°$ ⟶ greater "complexity" ⟶ greater disorder

compositional complexity

Substance (*state*)	$S° \left(\frac{J}{mol \cdot K}\right)$
$CO(g)$	197.7
$CO_2(g)$	213.8
$CaO(s)$	38.1
$CaCO_3(s)$	91.7

mass or no. of e⁻ complexity

Substance (*state*)	$S° \left(\frac{J}{mol \cdot K}\right)$
$H_2(g)$	130.7
$O_2(g)$	205.2
$I_2(g)$	260.7

phase complexity

Substance (*state*)	$S° \left(\frac{J}{mol \cdot K}\right)$
$H_2O(\ell)$	70.0
$H_2O(g)$	188.8
$Hg(\ell)$	75.9
$Hg(g)$	175.0
$I_2(s)$	116.2
$I_2(g)$	260.7

Entropy Changes

18.2

$$\overset{\text{reactants}}{aA + bB} \rightarrow \overset{\text{products}}{cC + dD}$$

n_R $\quad$ n_P

Calculating changes in enthalpy (ΔH^o_{rxn}) from tabulated ΔH^o_f values.

$$\Delta H^o_{rxn} = \underbrace{\sum n_P \cdot \Delta H^o_{f,P}}_{\text{products}} - \underbrace{\sum n_R \cdot \Delta H^o_{f,R}}_{\text{reactants}} = \left(c \cdot \Delta H^o_{f,C} + d \cdot \Delta H^o_{f,D}\right) - \left(a \cdot \Delta H^o_{f,A} + b \cdot \Delta H^o_{f,B}\right)$$

Calculating changes in entropy (ΔS^o_{rxn}) from tabulated S^o values.

$$\Delta S^o_{rxn} = \underbrace{\sum n_P \cdot S^o_P}_{\text{products}} - \underbrace{\sum n_R \cdot S^o_R}_{\text{reactants}} = \left(c \cdot S^o_C + d \cdot S^o_D\right) - \left(a \cdot S^o_A + b \cdot S^o_B\right)$$

IN-CLASS QUESTION 2

Learning Objective(s): 18.2, 18.4

Gaseous hydrogen and oxygen react to form water vapor.

a) Write a balanced chemical equation including phases.

$$2H_2(g) + 1O_2(g) \longrightarrow 2H_2O(g) \qquad \Delta n_{gas} = 2 - 3 = -1 < 0 \rightarrow \Delta S < 0$$

b) For this reaction, it is predicted that: $\Delta S^o_{rxn} < 0$ $\Delta S^o_{rxn} > 0$

c) Calculate the standard change in entropy for this reaction (ΔS^o_{rxn}) in J·mol^{-1}·K^{-1}.

$$\Delta S^\circ_{rxn} = \sum n_p \cdot S^\circ_P - \sum n_R \cdot S^\circ_R$$

$$= \left[2 \cdot S^\circ_{H_2O(g)}\right] - \left[2 \cdot S^\circ_{H_2(g)} + 1 \cdot S^\circ_{O_2(g)}\right]$$

$$= \left[(2)\left(188.8 \frac{J}{mol \cdot K}\right)\right] - \left[(2)\left(130.7 \frac{J}{mol \cdot K}\right) + (1)\left(205.2 \frac{J}{mol \cdot K}\right)\right]$$

$$\Delta S^\circ_{rxn} = -89.0 \frac{J}{mol \cdot K}$$

Substance (*state*)	$S^\circ \left(\frac{J}{mol \cdot K}\right)$
$H_2(g)$	130.7
$O_2(g)$	205.2
$H_2O(\ell)$	70.0
$H_2O(g)$	188.8

BONUS QUESTION

Assessed on: Summer 2023, Final Exam

Select all processes for which you expect $\Delta S < 0$.

- ❑ $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3\ H_2(g)$
- ❑ $C_2H_5OH(\ell) \rightarrow C_2H_5OH(s)$
- ❑ $N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH_3(g)$
- ❑ $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
- ❑ $Na_2CO_3(s) + H_2O(g) + CO_2(g) \rightarrow 2\ NaHCO_3(s)$
- ❑ $Al^{3+}(aq) + 3\ Cl^-(aq) \rightarrow AlCl_3(s)$